

Chan Nhu (Soleil) Nguyen Pham

soleilpham@gmail.com • (765) 806-0753 • [linkedin.com/in/chan-nhu-p-4876a127a](https://www.linkedin.com/in/chan-nhu-p-4876a127a) • github.com/soleilpham • soleilpham.github.io/source

EDUCATION

Purdue University — College of Science

B.S. in Computer Science (Machine Intelligence & Systems Software Concentration)

West Lafayette, IN

Aug 2024 – May 2027

- **Minor:** Mathematics
- **Relevant Coursework:** Artificial Intelligence, Algorithm Analysis, Systems Programming (C, Assembly), Data Structures & Algorithms, Computer Architecture, Competitive Programming (C++), Object-Oriented Programming (Java)

EXPERIENCE

Purdue University — College of Science

Undergraduate Researcher for PurSec, Snowball, and CoRAL Labs

West Lafayette, IN

Nov 2024 – Present

- **PurSec:** Conducted comprehensive literature reviews on Extended Reality (XR) privacy and security risks, leading discussions to guide the research direction; designed user surveys to assess security awareness of Large Language Models (LLMs) and Vision-Language Models (VLMs) integrated into AR applications.
- **Snowball:** Optimized data pipelines through rigorous code review, testing, and data interpolation to prepare datasets for model training; evaluated machine learning architectures for dietary classification using PPG signals and accelerometer data collected from custom smart-earring wearables developed for research studies.
- **CoRAL:** Analyzed state-of-the-art LLM and Machine Learning architectures to evaluate limitations and existing solutions for long-horizon robotic tasks; synthesized advanced techniques and architectural modifications to enhance sequential task planning and execution performance in robotic arms.

Purdue ACM SIGAPP

President & Technical Lead

West Lafayette, IN

Aug 2025 – Present

- Led backend development for two large-scale student-run software projects with ~20 contributors each, coordinating architecture decisions, sprint planning, and rigorous code reviews.
- Designed REST APIs, database schemas, and authentication flows for *BoilerBuzz*, a cross-platform mobile app backed by Node.js and PostgreSQL.
- Architected and developed a FastAPI-based grant management platform serving Purdue's graduate student community (10,000+ students), streamlining grant submission, review, approval, and reimbursement workflows.
- Implemented role-based access control with Keycloak, enabling secure multi-role authorization pipelines.

Reclub — Sports Technology Startup

Software Engineer Intern

Ho Chi Minh City, VN

May 2025 – Aug 2025

- Designed and deployed an automated video segmentation system that identifies pickleball rallies from raw match footage, reducing manual editing time from hours to minutes.
- Built and optimized a YOLOv10s-based, motion-aware classification pipeline, achieving 96% accuracy while targeting near-real-time, on-device inference.
- Implemented an end-to-end ML pipeline including large-scale data labeling (36K+ frames), training, and cloud deployment using Docker, GCS, and Vertex AI.

Purdue University — College of Liberal Arts

Web Developer

West Lafayette, IN

Jan 2026 – Present

- Maintained and extended university web services by developing reusable templates and responsive components using HTML, CSS, JavaScript, and Apache Velocity.
- Resolved IT support tickets involving content updates, accessibility improvements, and legacy system modernization.

Amazon Web Services (AWS) Youth Experience Program

Data Analytics Intern

Remote

Jul 2023 – Aug 2023

- Developed Python-based data processing pipelines using Pandas, NumPy, and scikit-learn to analyze wildfire risk through forest-type prediction, recognized as a top 5 intern project for technical quality and clarity of data analysis.
- Evaluated multiple ML models (Logistic Regression, Decision Trees, Random Forests, Neural Networks, KNN) to identify optimal features and improve predictive accuracy.

SKILLS

- **Languages:** Python, C++, C, Java, JavaScript, TypeScript, SQL, Assembly, Bash
- **Backend & Databases:** FastAPI, Node.js, PostgreSQL, MongoDB, Authentication, Keycloak, Multithreading
- **Data Science:** Pandas, NumPy, Data Analytics, Statistical Analysis, Model Evaluation
- **Cloud & DevOps:** Docker, Google Cloud Platform (GCS, Vertex AI), Git/GitHub